DSC 212 - Quiz 2 Prep

Nick Wendt

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Common Distributions

Table 1: Special Discrete Distributions

Notation and Parameters	Discrete pdf $f(x)$	Mean	Variance	MGF $M_X(t)$
Bernoulli				
$X \sim \text{Ber}(p)$	$p^x q^{1-x}$	p	pq	$pe^t + q$
$0 < p < 1$	$x = 0, 1$			
$q = 1 - p$				
Binomial				
$X \sim \text{Bin}(n, p)$	$\binom{n}{x} p^x q^{n-x}$	np	npq	$(pe^t + q)^n$
$0 < p < 1$	$x = 0, 1, \ldots, n$			
$q = 1 - p$				
Hypergeometric				
$X \sim \text{Hyp}(n, r, N)$	$\frac{\binom{r}{x}\binom{N-r}{n-x}}{\binom{N}{n}}$	$n\frac{r}{N}$	$n\frac{r}{N}\left(1 - \frac{r}{N}\right)\left(\frac{N-n}{N-1}\right)$	Not tractable
$n = 1, 2, \ldots, N$	$\max(0, r + n - N) \leq x \leq \min(r, n)$			
$r = 0, 1, \ldots, N$				
Geometric				
$X \sim \text{Geo}(p)$	pq^{x-1}	$\frac{1}{p}$	$\frac{q}{p^2}$	$\frac{pe^t}{1-qe^t}$
$0 < p < 1$	$x = 1, 2, \ldots$			
$q = 1 - p$				
Negative Binomial				
$X \sim \text{NB}(r, p)$	$\binom{x-1}{r-1} p^r q^{x-r}$	$\frac{r}{p}$	$\frac{rq}{p^2}$	$\left(\frac{pe^t}{1-qe^t}\right)^r$
$0 < p < 1$	$x = r, r + 1, \ldots$			
$r = 1, 2, \ldots$				
Poisson				
$X \sim \text{Poi}(\lambda)$	$\frac{e^{-\lambda}\lambda^x}{x!}$	λ	λ	$e^{\lambda(e^t-1)}$
$\lambda > 0$	$x = 0, 1, \ldots$			
Discrete Uniform				
$X \sim \text{DU}(N)$	$\frac{1}{N}$	$\frac{N+1}{2}$	$\frac{N^2-1}{12}$	$\frac{1}{N}\frac{e^t - e^{(N+1)t}}{1-e^t}$
$N = 1, 2, \ldots$	$x = 1, 2, \ldots, N$			

Table 2: Special Continuous Distributions

Notation and Parameters	Continuous pdf $f(x)$	Mean	Variance	MGF $M_X(t)$
Uniform				
$X \sim \mathrm{U}(a, b)$	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{e^{bt}-e^{at}}{(b-a)t}$
$a < b$	$a < x < b$			
Normal				
$X \sim \mathrm{N}(\mu, \sigma^2)$	$\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$	μ	σ^2	$e^{\mu t+\sigma^2 t^2/2}$
$-\infty < \mu < \infty$	$-\infty < x < \infty$			
$\sigma^2 > 0$				
Gamma				
$X \sim \mathrm{Gam}(r, \lambda)$	$\frac{\lambda^r}{\Gamma(r)} x^{r-1} e^{-\lambda x}$	$\frac{r}{\lambda}$	$\frac{r}{\lambda^2}$	$\left(\frac{1}{1-\frac{t}{\lambda}}\right)^r$
$\lambda > 0$	$x > 0$			
$r > 0$				
Exponential				
$X \sim \mathrm{Exp}(\lambda)$	$\lambda e^{-\lambda x}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$\frac{1}{1-\frac{t}{\lambda}}$
$\lambda > 0$	$x > 0$			

Key Formulas

Valid joint pdf / pmf. $f(x,y) \geq 0$ everywhere, and it sums or integrates to 1 over its support:

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(x,y)\,dx\,dy = 1 \qquad \text{or} \qquad \sum_x \sum_y p(x,y) = 1.$$

Probability of a region. For a region A in the xy-plane,

$$P\big((X,Y) \in A\big) = \iint_A f(x,y)\,dx\,dy.$$

Marginals. Integrate or sum out the other variable:

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)\,dy, \qquad f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\,dx, \qquad p_X(x) = \sum_y p(x,y), \qquad p_Y(y) = \sum_x p(x,y).$$

Conditionals. The joint divided by the marginal of the variable you condition on:

$$f_{Y|X}(y \mid x) = \frac{f(x,y)}{f_X(x)}, \qquad f_{X|Y}(x \mid y) = \frac{f(x,y)}{f_Y(y)}, \qquad f_X(x) > 0,\ f_Y(y) > 0.$$

Multiplication rule. Rearranging the conditional:

$$f(x,y) = f_{Y|X}(y \mid x)\, f_X(x) = f_{X|Y}(x \mid y)\, f_Y(y).$$

Independence. X and Y are independent if and only if the joint pdf equals the product of its marginals for *every* pair (x,y):

$$f(x,y) = f_X(x)\, f_Y(y) \qquad \text{or} \qquad p(x,y) = p_X(x)\, p_Y(y).$$

Equivalently, $f_{Y|X}(y \mid x) = f_Y(y)$: knowing X tells you nothing about Y.

Expectation. For any function g,

$$E[g(X,Y)] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g(x,y)\, f(x,y)\,dx\,dy.$$

If X and Y are independent, then $E[XY] = E[X]\,E[Y]$.

Practice Problems

1. Let X represent the number of heads you observe from tossing a fair coin 100 times. What distribution does X follow? Calculate the probability of getting exactly 50 heads from the 100 tosses.

1a. Now repeat the experiment, tossing the same coin until you observe 50 heads, and let X represent the number of tosses it takes. What distribution does X follow? Calculate the probability that the 50th head occurs on exactly the 100th toss.

2. Let X and Y be continuous random variables with joint density

$$f(x,y) = \begin{cases} c\,xy^2, & 0 \le x \le 2,\ 0 \le y \le 3 \\ 0, & \text{otherwise.} \end{cases}$$

Find the value of the constant c that makes $f(x,y)$ a valid joint pdf.

3. Let X and Y be continuous random variables with joint density

$$f(x,y) = \begin{cases} x+y, & 0 \le x \le 1,\ 0 \le y \le 1 \\ 0, & \text{otherwise.} \end{cases}$$

Find the conditional density $f_{Y|X}(y \mid x)$, then use it to calculate $P\left(Y \le \frac{1}{2} \,\middle|\, X = \frac{1}{2}\right)$.

4. Let X and Y be discrete random variables with the joint pmf $p(x,y)$ given in the table below.

$p(x,y)$	$Y=0$	$Y=1$
$X=0$	0.08	0.12
$X=1$	0.20	0.30
$X=2$	0.12	0.18

Find the marginal pmfs $p_X(x)$ and $p_Y(y)$. Are X and Y independent?

5. Let X and Y be continuous random variables with joint density

$$f(x,y) = \begin{cases} \frac{3}{2}x + \frac{1}{2}y, & 0 \le x \le 1,\ 0 \le y \le 1 \\ 0, & \text{otherwise.} \end{cases}$$

Calculate $E[X]$, $E[Y]$, and $E[XY]$. Is $E[XY] = E[X]\,E[Y]$?

Answers

1. Each toss is an independent Bernoulli trial with success (heads) probability $p = \frac{1}{2}$, and X counts the successes in a fixed number $n = 100$ of trials, so $X \sim \text{Bin}(n = 100,\ p = \frac{1}{2})$ with

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, 1, \ldots, n.$$

Plugging in $k = 50$:

$$\begin{aligned} P(X = 50) &= \binom{100}{50} \left(\tfrac{1}{2}\right)^{50} \left(\tfrac{1}{2}\right)^{50} \\ &= \binom{100}{50} \left(\tfrac{1}{2}\right)^{100} \\ &\approx 0.0796 \end{aligned}$$

1a. Now the number of successes is fixed at $r = 50$ and X counts the trials needed to reach them, so $X \sim \text{NB}(r = 50,\ p = \frac{1}{2})$ with

$$P(X = k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \qquad k = r, r+1, \ldots$$

For the 50th head to land on exactly the 100th toss, the first 99 tosses must contain exactly 49 heads, and the 100th toss must be a head. Plugging in $k = 100$:

$$\begin{aligned} P(X = 100) &= \binom{99}{49} \left(\tfrac{1}{2}\right)^{50} \left(\tfrac{1}{2}\right)^{50} \\ &= \binom{99}{49} \left(\tfrac{1}{2}\right)^{100} \\ &\approx 0.0398 \end{aligned}$$

This is exactly half of the answer to 1, since $\binom{99}{49} = \frac{50}{100}\binom{100}{50}$: of all the sequences with 50 heads in 100 tosses, exactly half end in a head.

2. For $f(x, y)$ to be a valid joint pdf, it must be nonnegative and integrate to 1 over its support:

$$\int_0^3 \int_0^2 c\, xy^2 \, dx \, dy = 1.$$

Evaluating the inner integral with respect to x (treating y as a constant):

$$\int_0^2 c\, xy^2 \, dx = c\, y^2 \left[\frac{x^2}{2}\right]_0^2 = 2c\, y^2.$$

Then the outer integral with respect to y:

$$\int_0^3 2c\, y^2 \, dy = 2c \left[\frac{y^3}{3}\right]_0^3 = 2c \cdot 9 = 18c.$$

Setting $18c = 1$ gives

$$c = \frac{1}{18}.$$

Since $xy^2 \geq 0$ on the support, $f(x, y) \geq 0$ for this c, so f is a valid joint pdf.

3. The conditional density is the joint density divided by the marginal of the variable we condition on:

$$f_{Y|X}(y \mid x) = \frac{f(x,y)}{f_X(x)}.$$

First find the marginal of X by integrating out y:

$$f_X(x) = \int_0^1 (x+y)\,dy = \left[xy + \frac{y^2}{2}\right]_0^1 = x + \frac{1}{2}, \qquad 0 \le x \le 1.$$

Dividing the joint by the marginal:

$$f_{Y|X}(y \mid x) = \frac{x+y}{x+\frac{1}{2}}, \qquad 0 \le y \le 1.$$

At $x = \frac{1}{2}$ the denominator is $\frac{1}{2} + \frac{1}{2} = 1$, so $f_{Y|X}\left(y \mid \frac{1}{2}\right) = \frac{1}{2} + y$. Then

$$P\left(Y \le \tfrac{1}{2} \,\middle|\, X = \tfrac{1}{2}\right) = \int_0^{1/2} \left(\frac{1}{2} + y\right) dy = \left[\frac{y}{2} + \frac{y^2}{2}\right]_0^{1/2} = \frac{1}{4} + \frac{1}{8} = \frac{3}{8}.$$

4. Find each marginal by summing the joint pmf across a row (for X) or down a column (for Y):

$p(x,y)$	$Y=0$	$Y=1$	$p_X(x)$
$X=0$	0.08	0.12	0.20
$X=1$	0.20	0.30	0.50
$X=2$	0.12	0.18	0.30
$p_Y(y)$	0.40	0.60	1

So $p_X(0) = 0.2$, $p_X(1) = 0.5$, $p_X(2) = 0.3$ and $p_Y(0) = 0.4$, $p_Y(1) = 0.6$.

X and Y are independent if $p(x,y) = p_X(x)\,p_Y(y)$ for every cell. Checking all six:

$$\begin{aligned}
p_X(0)\,p_Y(0) &= (0.2)(0.4) = 0.08 = p(0,0), & p_X(0)\,p_Y(1) &= (0.2)(0.6) = 0.12 = p(0,1),\\
p_X(1)\,p_Y(0) &= (0.5)(0.4) = 0.20 = p(1,0), & p_X(1)\,p_Y(1) &= (0.5)(0.6) = 0.30 = p(1,1),\\
p_X(2)\,p_Y(0) &= (0.3)(0.4) = 0.12 = p(2,0), & p_X(2)\,p_Y(1) &= (0.3)(0.6) = 0.18 = p(2,1).
\end{aligned}$$

Every cell matches, so X and Y are independent. Equivalently, each row of the table is proportional to $(0.4, 0.6)$, so the conditional distribution of Y given $X = x$ is the same for every x.

5. For $E[X]$:

$$E[X] = \int_0^1 \int_0^1 x\left(\frac{3}{2}x + \frac{1}{2}y\right) dy\, dx = \int_0^1 \int_0^1 \left(\frac{3}{2}x^2 + \frac{1}{2}xy\right) dy\, dx.$$

Inner integral with respect to y:

$$\int_0^1 \left(\frac{3}{2}x^2 + \frac{1}{2}xy\right) dy = \left[\frac{3}{2}x^2 y + \frac{1}{4}xy^2\right]_0^1 = \frac{3}{2}x^2 + \frac{1}{4}x.$$

Outer integral with respect to x:

$$E[X] = \int_0^1 \left(\frac{3}{2}x^2 + \frac{1}{4}x\right) dx = \left[\frac{1}{2}x^3 + \frac{1}{8}x^2\right]_0^1 = \frac{1}{2} + \frac{1}{8} = \frac{5}{8}.$$

For $E[Y]$:

$$E[Y] = \int_0^1 \int_0^1 y\left(\frac{3}{2}x + \frac{1}{2}y\right) dx\, dy = \int_0^1 \int_0^1 \left(\frac{3}{2}xy + \frac{1}{2}y^2\right) dx\, dy.$$

Inner integral with respect to x:

$$\int_0^1 \left(\frac{3}{2}xy + \frac{1}{2}y^2\right) dx = \left[\frac{3}{4}x^2 y + \frac{1}{2}xy^2\right]_0^1 = \frac{3}{4}y + \frac{1}{2}y^2.$$

Outer integral with respect to y:

$$E[Y] = \int_0^1 \left(\frac{3}{4}y + \frac{1}{2}y^2\right) dy = \left[\frac{3}{8}y^2 + \frac{1}{6}y^3\right]_0^1 = \frac{3}{8} + \frac{1}{6} = \frac{13}{24}.$$

For $E[XY]$:

$$E[XY] = \int_0^1 \int_0^1 xy\left(\frac{3}{2}x + \frac{1}{2}y\right) dx\, dy = \int_0^1 \int_0^1 \left(\frac{3}{2}x^2 y + \frac{1}{2}xy^2\right) dx\, dy.$$

Inner integral with respect to x:

$$\int_0^1 \left(\frac{3}{2}x^2 y + \frac{1}{2}xy^2\right) dx = \left[\frac{1}{2}x^3 y + \frac{1}{4}x^2 y^2\right]_0^1 = \frac{1}{2}y + \frac{1}{4}y^2.$$

Outer integral with respect to y:

$$E[XY] = \int_0^1 \left(\frac{1}{2}y + \frac{1}{4}y^2\right) dy = \left[\frac{1}{4}y^2 + \frac{1}{12}y^3\right]_0^1 = \frac{1}{4} + \frac{1}{12} = \frac{1}{3}.$$

Comparing with the product of the means:

$$E[X]\,E[Y] = \frac{5}{8} \cdot \frac{13}{24} = \frac{65}{192} \neq \frac{64}{192} = \frac{1}{3} = E[XY].$$

So $E[XY] \neq E[X]\,E[Y]$, which means X and Y are *not* independent. This is the contrapositive of the direction that always holds:

$$X, Y \text{ independent} \implies E[XY] = E[X]\,E[Y], \qquad \text{so} \qquad E[XY] \neq E[X]\,E[Y] \implies X, Y \text{ not independent.}$$

Note: $E[XY] = E[X]\,E[Y]$ on its own does not prove independence.